

STRING PCS PROTOCOL

(V1.2.6)

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PCS BD, DELTA ELECTRONICS, INC.

Protocol Modification Notes

Version	Date	Author	Notes
V1.0.0	2022/11/21	Zhiguang.Gao	Draft version
V1.0.1	2023/1/29	Zhiguang.Gao	Fixed some description errors.
V1.1.0	2023/4/29	Zhiguang.Gao	1. The register address mapping is optimized and the register address in the protocol is redefined. 2. Updated the text description of some register functions in the documentation.
V1.2.0	2023/6/9	Zhiguang.Gao	1. The register mapping and description of protocol are re-modified according to the dual-master communication architecture
V1.2.1	2023/8/24	Zhiguang.Gao	1. Fixed some description error. 2. Add register d30111 to display Insulation resistance value.
V1.2.2	2023/9/25	Zhiguang.Gao	1. Add a description explaining the diagram “String PCS System Communication Bus Connection Diagram”.
V1.2.3	2024/10/12	Qiguo.Shi	1. Fixed some error, added some description of Faults.
V1.2.4	2025/1/6	Zhiguang Gao	1. Fixed some error, added some description. 2. Add register 42206 and 42207 to be used for battery single cluster controllable
V1.2.5	2025/2/8	Zhiguang Gao	1. Fixed register d30103 and d30104 definition
V1.2.6	2025/3/26	Zhiguang Gao	1. Add some descriptions about the Power Unit 2. Add power unit 1#12 status registers d30203-d30214. 3. Split original chapter6.2.2 Modules Monitoring into chapter6.2.2 Units Monitoring and chapter 6.2.3 Modules Monitoring 4. Modify the descriptions about registers d30354-d30401 of chapter6.2.2 5. Add a note about POW_MOD#n_MEAS_PF from modules, when real apparent power is less than 1kVA, this value will be showed as 1001, which means this value is invalid.

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1. Purpose

This document is written for user to better understand the system architecture of String PCS and definition of registers, and makes users understand of how to use and configure these registers flexibly on String PCS system. In this protocol, String PCS 2580 model including 12 power modules is used as an example.

2. Abbreviations

ID	Address Of Register Map
DSP	Digital Signal Process Unit
OA	Online Access, if it is YES, which means that the corresponding register could be accessed in any state of PCS
PCC	Point of Common Coupling
PCS	Power Conditioning System
R	Read Only
RW	Readable / Writable
W	Write Only
CC	Center Controller
SC	System Controller
MC	Module Controller

Note: The Protocol file is constantly improved and increased with the product update and the increase of new functions. Of course, there are inevitably some problems in the description of the document that need to be constantly revised. These changes or improved contents may not be timely updated to the customer.

3. System Architecture

3.1. Classic Application Connection

In order to describe String PCS operation, we define a general system architecture as follows. And the system mentioned in this document would be the same as follows. Notes would be added when different appears.

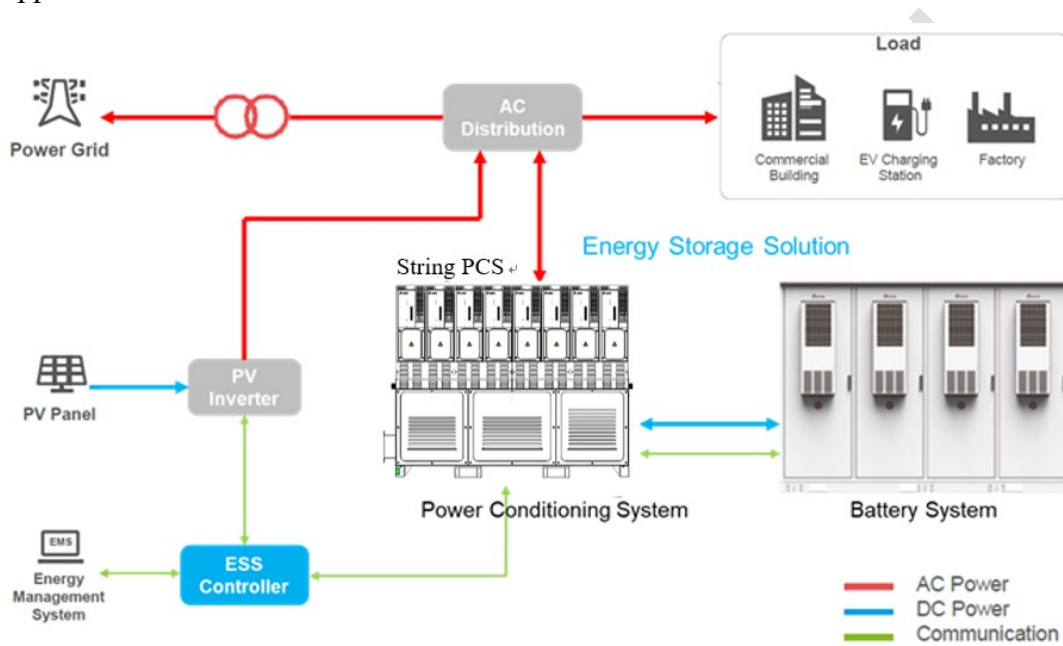


Figure 3-1 Schematic Diagram of String PCS System Application

3.2. String PCS System Hierarchy

The control architecture of the string system is divided into two layers, namely system level, power module level. The system level is managed by a high-performance ARM processor (center controller, CC), which is responsible for internal and external data interaction and communication, and a high-performance DSP controller (system controller, SC) for calculation, power module management and state machine control. The power unit level is a virtual power management unit, which is integrated in system controller as a function and manage multi power modules parallel control. The power layer mainly realizes the power conversion function, that is, the execution component of the power conversion, which contains a high-performance DSP controller (power module controller, MC) and the power component, such as IGBTs, and the control loop such as the power control loop and the current loop are realized in the power module controller .

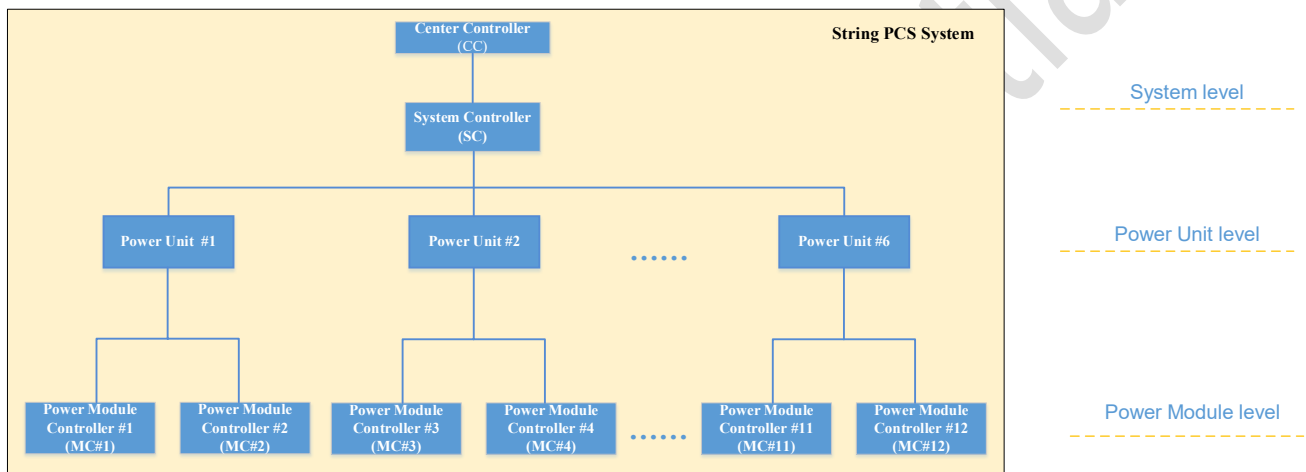


Figure 3-2 String PCS System Hierarchy Diagram (Grouping Type F as an example, see [Table 1](#) for details)

3.3. Communication Connection

The following is the internal and external connection diagram of communication bus, and it clearly show connection relationship between the main micro controllers and power modules.

The String System includes a sets of system control unit (SCU), which includes two core controllers, namely Center controller (CC) and System controller (SC), the user can refer to Section 3.2 for more information.

SCU can manages up to 12 power modules simultaneously. There are optical fiber fast communication and Ethernet conventional communication between SCU and each power module. The switch implements the information interaction between the center controller, system controller, and power module controllers. They are a small local area network.

Fast instructions are transmitted internally through the optical fiber bus. Fast instructions include ON and OFF, reactive/active power command, etc. Slow read and write operations are transmitted over Ethernet cable and include parameter configuration, operation mode configuration, and log reading etc.

The BMS controller or EMS controller can be directly connected to the SCU via Ethernet cables to control string PCS system based on the Modbus TCP protocol.

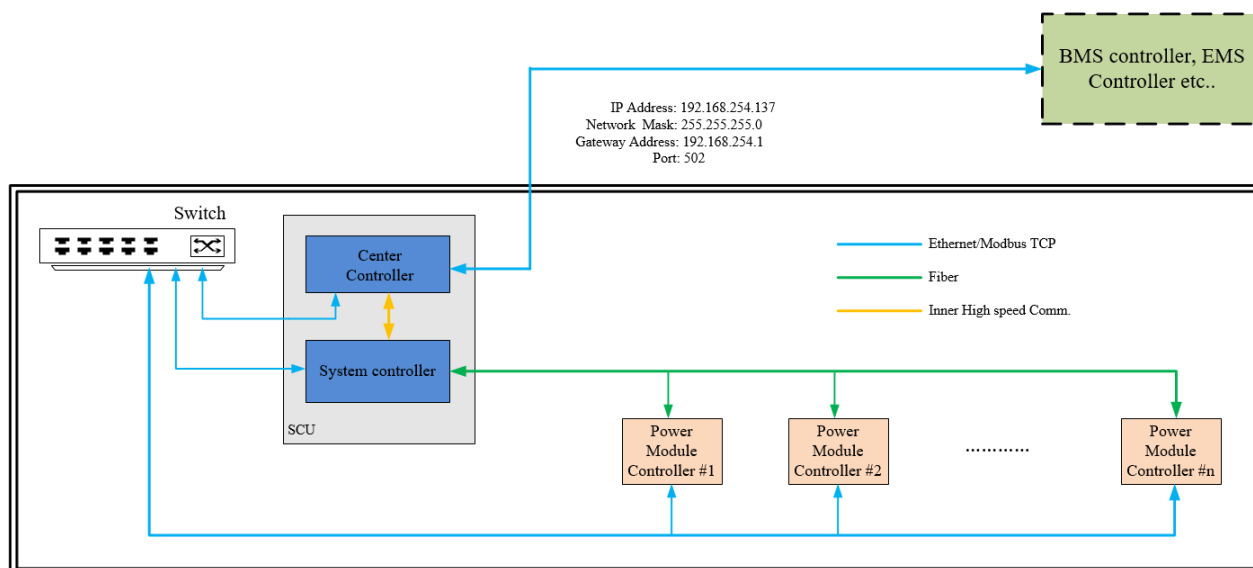


Figure 3-3 String PCS System Communication Bus Connection Diagram (n=3 to 12)

Notes:

1. IP address 192.168.1.* is used to internal devices of string PCS system. The IP address of the center controller for external communication with EMS controller does not use the 192.168.1.* network segment due to the hardware limitation.
2. The communication protocol between BMS and PCS is different due to different BMS brands and need to be customized development.

3.4. Read/Write Register

When the upper stream controller successfully connects to the center controller of String PCS through Ethernet cable with Modbus TCP Protocol, the user can access all registers described in following chapters. Data segment format is shown as below:

Transaction process identifier	Protocol identifier	Data length	Address	Function code	Byte quantity	Data
2*byte	2*byte	2*byte	1*byte	1*byte	1*byte	N*byte

1. Transaction process identifier (byte0, byte1) --- 00 00
2. Protocol identifier (byte2, byte3) --- 00 00
3. Length segment (byte4, byte5) --- it means the overall data length (address + function code + data) following.

4. Address (byte6) --- it means the address of the guest computer being addressed; this protocol specified the address of guest computer is 0x01h.
5. Function code (byte7) --- order the guest computer being addressed guest to finish required action. Function code as noted in the table below.
6. Data (byte8~xx) --- data field, includes data guest terminal needs to finish some specified function, and includes uploaded data during servicing.

Function code introduction supported by center controller:

Modbus TCP	Description
0x03	Read holding registers
0x06	Write single register
0x10	Write multiple registers

4. Configuration

Note: Only need to separately write configuration parameters to the center controller of system, then the center controller will copy it to each power module in system through internal Ethernet (Modbus TCP) bus.

4.1. System Specification

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default	OA
A474	42100	Uint16	SYS_RATING_AC_POWRE	Rating Power	0.1kVA	R	2150	25800	25800	No
A475	42101	Uint16	SYS_RATING_AC_VOLT	Rating AC Voltage	0.1V	R	3800	6900	6900	No
A476	42102	Uint16	SYS_RATING_AC_FREQ	Rating AC Frequency	0.01Hz	R	4500	7000	6000	No
A477	42103	Uint16	SYS_RATING_AC_CURR	Rating AC Current	0.1A	R	0	21600	21600	No
A478	42104	Uint16	Reserved	/	/	/	/	/	/	/
A479	42105	Uint16	SYS_ALTITUDE_SETTING	Altitude Setting	m	RW	0	10000	0	No

Note:

- 1) The default values in the above table will be different depending on the string PCS model.

4.2. Power Unit/Module Configuration

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default	OA																																																
A4DE	42206	Uint16	SYS_GROUP_TYPE_SETTING	System Grouping Type, please refer to Table 1.	NA	R	0	6	6	No																																																
A4DF	42207	Uint16	SYS_UNIT_ALONE_ENABLE	Sub-System A master-slave type. 0: Master-slave mode String PCS system receives total power command from EMS controller. The active power and/or reactive power command should be written to reg.42403 and/or 42404 of. The ON/OFF command should be written to reg. 42400. 1: Dual master mode Each center controller separately receives power command from EMS controller. The active power and/or reactive power commands should be separately written to reg.42403 to 42404 of each center controller. The ON/OFF command should be written to reg. 42400.	NA	RW	0	1	0	No																																																
A4E2	42210	Uint16	SYS_MOD_ENABLE_CONF	System power module enables. See the below table for the definition of each bit.	NA	RW	0	4095	4095	No																																																
<table><tr><th>bit 15</th><th>bit 14</th><th>bit 13</th><th>bit 12</th><th>bit 11</th><th>bit 10</th><th>bit 9</th><th>bit 8</th></tr><tr><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>Power module #12</td><td>Power module #11</td><td>Power module #10</td><td>Power module #9</td></tr><tr><td>RW-0h</td><td>RW-0h</td><td>RW-0h</td><td>RW-0h</td><td>RW-0h</td><td>RW-1h</td><td>RW-1h</td><td>RW-1h</td></tr><tr><th>bit 7</th><th>bit 6</th><th>bit 5</th><th>bit 4</th><th>bit 3</th><th>bit 2</th><th>bit 1</th><th>bit0</th></tr><tr><td>Power module #8</td><td>Power module #7</td><td>Power module #6</td><td>Power module #5</td><td>Power module #4</td><td>Power module #3</td><td>Power module #2</td><td>Power module #1</td></tr><tr><td>RW-1h</td><td>RW-1h</td><td>RW-1h</td><td>RW-1h</td><td>RW-1h</td><td>RW-1h</td><td>RW-1h</td><td>RW-0h</td></tr></table>											bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8	NA	NA	NA	NA	Power module #12	Power module #11	Power module #10	Power module #9	RW-0h	RW-0h	RW-0h	RW-0h	RW-0h	RW-1h	RW-1h	RW-1h	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0	Power module #8	Power module #7	Power module #6	Power module #5	Power module #4	Power module #3	Power module #2	Power module #1	RW-1h	RW-1h	RW-1h	RW-1h	RW-1h	RW-1h	RW-1h	RW-0h
bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8																																																			
NA	NA	NA	NA	Power module #12	Power module #11	Power module #10	Power module #9																																																			
RW-0h	RW-0h	RW-0h	RW-0h	RW-0h	RW-1h	RW-1h	RW-1h																																																			
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0																																																			
Power module #8	Power module #7	Power module #6	Power module #5	Power module #4	Power module #3	Power module #2	Power module #1																																																			
RW-1h	RW-1h	RW-1h	RW-1h	RW-1h	RW-1h	RW-1h	RW-0h																																																			

Note:

1. Reg. 42207 should be changed when String PCS is in standby state.
2. Reg. 42206 is set at the factory and cannot be modified.

Table 1 String PCS grouping type configuration (Reg. 42206)

Value	Group type	Units number	Power module /per unit	Remark
0	Type A	1	12	For Type E, total power module number is 10. The others are defined based on total number of power module being 12. To better understand what is grouping types, an example of Type F is provided, please refer to the below Figure 4-1.
1	Type B	2	6	
2	Type C	3	4	
3	Type D	4	3	
4	Type E	5	2	
5	Type F	6	2	
6	Type G	12	1	

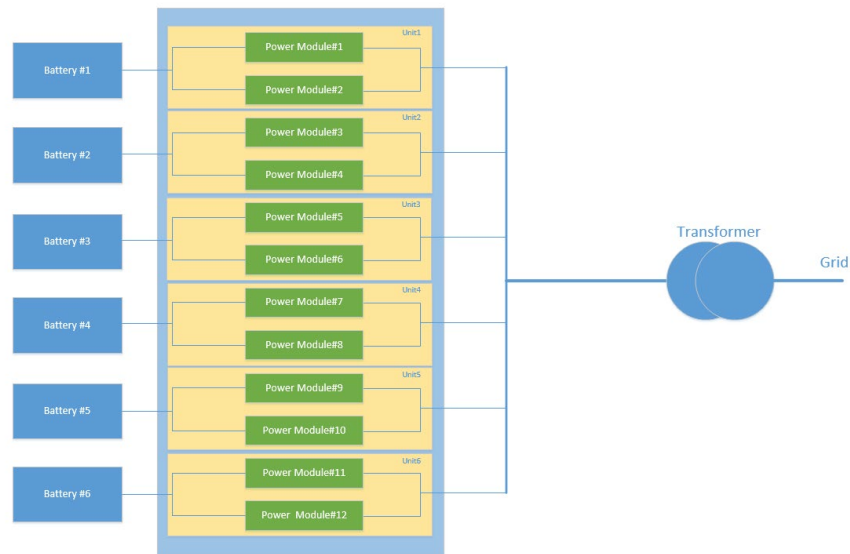


Figure 4-1 Example drawing for Type F Configuration

4.3. Grid Code

The register parameters will vary depending on the register 43000(Grid code type). The parameters in the table below are the requirements of IEEE1547-2018, and all grid code parameters can be accessed online.

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4.3.1. Basic Parameter

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default	OA
A7F8	43000	Uint16	SYS_GRID_CODE	Grid code type: 0: None 1: IEEE1547-2018	NA	RW	0	1	1	No
A7F9	43001	Uint16	GC_Q_MODE_SEL	Reactive Power Mode Selection 0: Power Set (Demand response) 1: SPF 2: Reserved 3: Volt-Var 4: Watt-Var Other: NA	NA	RW	0	4	0	Yes
A7FA	43002	Uint16	GC_PQ_PRIORITY	Priority of active power / reactive power (Grid-following mode) 0: Reactive first ; 1: Active first;	NA	RW	0	1	0	Yes

4.3.2. Normal/Soft-Start Ramp Rates

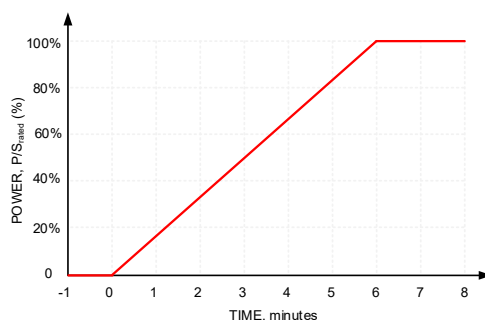


Figure 4-2 Example: Power Ramp Rate (register[43004]=33)

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A7FB	43003	Uint16	GC_NORM_RR_EN	Normal Ramp Rate Enable at On-grid mode Switch: =0: Disable, Scale: =1: Enable with 0.01% Scale =2: Enable with 0.1% Scale =3: Enable with 1% Scale	NA	RW	0	3	1
A7FC	43004	Uint16	GC_NORM_RR_VAL	Normal Ramp Rate Value Unit: 0.01%/s, 0.1%/s, 1%/s; Depending on enable register.[43003].	0.01/0.1/1%/s	RW	1	10000	33
A7FD	43005	Uint16	GC_NORM_SOFT_RR_EN	Normal Soft Start Ramp Rate Enable at Grid-following mode Switch: =0: Disable, Scale: =1: Enable with 0.01% Scale =2: Enable with 0.1% Scale =3: Enable with 1% Scale	NA	RW	0	3	0
A7FE	43006	Uint16	GC_NORM_SOFT_RR_VAL	Normal Soft Start Ramp Rate Value Unit: 0.01%/s, 0.1%/s, 1%/s; Depending on enable register. [43005]	0.01/0.1/1%/s	RW	1	10000	33

Note:

- For Normal ramp rate function, if the register [43004] is set to 33 and [43003] is set to 1, means that the ramp rate is 0.33%/s. if the normal ramp rate function is disabled, the power response time (0 to 100%Prated) is about 10ms.

4.3.3. High/Low Voltage Abnormal Protection

Table 13—DER response (shall trip) to abnormal voltages for DER of abnormal operating performance Category III (see Figure H.9)

Shall trip function	Shall trip—Category III			
	Default settings ^a		Ranges of allowable settings ^b	
	Voltage (p.u. of nominal voltage)	Clearing time (s)	Voltage (p.u. of nominal voltage)	Clearing time (s)
OV2	1.20	0.16	fixed at 1.20	fixed at 0.16
OV1	1.10	13.0	1.10 – 1.20	1.0 – 13.0
UV1	0.88	21.0	0.0 – 0.88	21.0 – 50.0
UV2	0.50	2.0	0.0 – 0.50	2.0 – 21.0

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A804	43012	Uint16	GC_AC_VOLT_LEVEL_EN	Over/Under Voltage Level Configuration OVP: 2 Levels. UVP: 3 Levels. Protection (PROT) Switch Enable bitx = 1: Check enable, bitx = 0: Check disable bit4: Over Level 2 enable bit3: Over Level 1 enable bit2: Under Level 1 enable bit1: Under Level 2 enable bit0: Under Level 3 enable	NA	RW	0	55	51
A805	43013	Uint16	GC_AC_VOLT_TIME_SCALE	Over/Under Voltage Time Scale Configuration OVP: 2 Levels. UVP: 3 Levels. Protection(PROT)/Ride Through(VRT) Time Scale bitx = 1: 0.1s(100ms), bitx = 0: 0.01s(10ms) bit4: Over Level 2 scale bit3: Over Level 1 scale bit2: Under Level 1 scale bit1: Under Level 2 scale bit0: Under Level 3 scale	NA	RW	0	55	0
A806	43014	Uint16	GC_AC_OVP_HV2	Over Voltage Protection High Value-Level 2 Configured by bit4 of Reg 43012	0.1%	RW	1200	1200	1200
A807	43015	Uint16	GC_AC_OVP_HV1	Over Voltage Protection High Value-Level 1 Configured by bit3 of Reg 43012	0.1%	RW	1100	1200	1100
A808	43016	Uint16	GC_AC_UVP_LV1	Under Voltage Protection Low Value-Level 1 Configured by bit2 of Reg 43012	0.1%	RW	50	880	880
A809	43017	Uint16	GC_AC_UVP_LV2	Under Voltage Protection Low Value-Level 2 Configured by bit1 of Reg 43012	0.1%	RW	50	500	500
A80A	43018	Uint16	GC_AC_UVP_LV3	Under Voltage Protection Low Value-Level 3 Configured by bit0 of Reg 43012	0.1%	RW	50	500	500
A80B	43019	Uint16	GC_AC_OVP_TIME_HV2	Over Voltage Protection Time High Value-Level 2 Configured by bit4 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	16	16
A80C	43020	Uint16	GC_AC_OVP_TIME_HV1	Over Voltage Protection Time High Value-Level 1 Configured by bit3 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	100	1300	1300
A80D	43021	Uint16	GC_AC_UVP_TIME_LV1	Under Voltage Protection Time Low Value-Level 1 Configured by bit2 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	200	5000	2100

A80E	43022	Uint16	GC_AC_UVP_TIME_LV2	Under Voltage Protection Time Low Value-Level 2 Configured by bit1 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	2100	200
A80F	43023	Uint16	GC_AC_UVP_TIME_LV3	Under Voltage Protection Time Low Value-Level 3 Configured by bit0 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	2100	200

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4.3.4. High/Low Frequency Abnormal Protection

Table 18—DER response (shall trip) to abnormal frequencies for DER of abnormal operating performance Category I, Category II, and Category III (see Figure H.10)

Shall trip function	Default settings ^a		Ranges of allowable settings ^b	
	Frequency ^c (Hz)	Clearing time (s)	Frequency (Hz)	Clearing time (s)
OF2	62.0	0.16	61.8–66.0	0.16–1 000.0
OF1	61.2	300.0	61.0–66.0	180.0–1 000.0
UF1	58.5	300.0 ^c	50.0–59.0	180.0–1 000
UF2	56.5	0.16	50.0–57.0	0.16–1 000

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A810	43024	Uint16	GC_AC_FREQ_LEVEL_EN	Over/Under Frequency Level Configuration OFP: 2 Levels. UFP: 2 Levels. Protection (PROT) Switch Enable bitx = 1: Check enable, bitx = 0: Check disable bit3: Over Level 2 enable bit2: Over Level 1 enable bit1: Under Level 1 enable bit0: Under Level 2 enable	NA	RW	0	51	51
A811	43025	Uint16	GC_AC_FREQ_TIME_SCALE	Over/Under Frequency Time Scale Configuration OVP: 2 Levels. UVP: 2 Levels. Protection(PROT)/Ride Through(FRT) Time Scale bitx = 1: 0.1s(100ms), bitx = 0: 0.01s(10ms) bit3: Over Level 2 scale bit2: Over Level 1 scale bit1: Under Level 1 scale bit0: Under Level 2 scale	NA	RW	0	51	17
A812	43026	Uint16	GC_AC_OFP_HF2	Over Frequency Protection High Value-Level 2 Config by bit3 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	6180	6600	6200
A813	43027	Uint16	GC_AC_OFP_HF1	Over Frequency Protection High Value-Level 1 Config by bit2 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	6100	6600	6120
A814	43028	Uint16	GC_AC_UFP_LF1	Under Frequency Protection Low Value-Level 1 Config by bit1 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	5000	5900	5850
A815	43029	Uint16	GC_AC_UFP_LF2	Under Frequency Protection Low Value-Level 2 Config by bit0 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	5000	5700	5650
A816	43030	Uint16	GC_AC_OFP_TIME_HF2	Over Frequency Protection Time High Value-Level 2 Config by bit3 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	10000	16
A817	43031	Uint16	GC_AC_OFP_TIME_HF1	Over Frequency Protection Time High Value-Level 1 Config by bit2 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	1800	10000	3000
A818	43032	Uint16	GC_AC_UFP_TIME_LF1	Under Frequency Protection Time Low Value-Level 1 Config by bit1 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	1800	10000	3000
A819	43033	Uint16	GC_AC_UFP_TIME_LF2	Under Frequency Protection Time Low Value-Level 2 Config by bit0 of Reg 43025	0.1/0.01s	RW	16	10000	16

				Unit: 0.01s(10ms), 0.1s(100ms)					
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4.3.5. Reconnection

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A81A	43034	Uint16	GC_AUTO_RECONNECT	Auto Reconnection Function Enable 0: Disable 1: Enable	NA	RW	0	1	1
A81B	43035	Uint16	GC_AC_RECONNECT_HV	Auto Reconnection Voltage Set High Unit: 0.1%	0.1%	RW	1050	1060	1050
A81C	43036	Uint16	GC_AC_RECONNECT_LV	Auto Reconnection Voltage Set Low Unit: 0.1%	0.1%	RW	880	950	917
A81D	43037	Uint16	GC_AC_RECONNECT_HF	Auto Reconnection Frequency Set High Unit: 0.01Hz	0.01Hz	RW	6010	6100	6010
A81E	43038	Uint16	GC_AC_RECONNECT_LF	Auto Reconnection Frequency Set Low Unit: 0.01Hz	0.01Hz	RW	5900	5990	5950
A81F	43039	Uint16	GC_AC_RECONNECT_TIME	Auto Reconnection Time Set Delay Unit: s	s	RW	1	600	300

4.3.6. SPF function

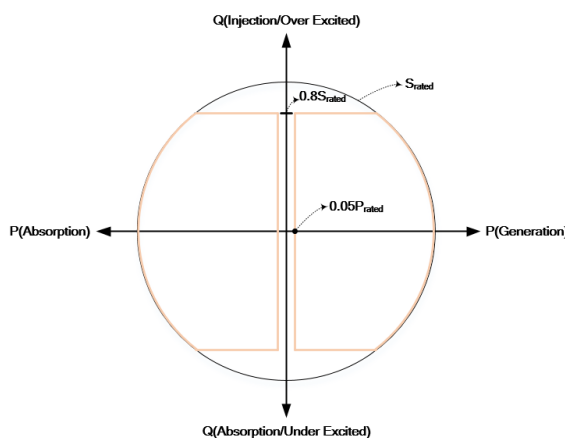


Figure 4-3 SPF Control Mode Limit Area

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A820	43040	Uint16	GC_AF_Q_ABS_MAX	Advance Function Q Max Set (Abs Value)	0.1%	RW	0	1000	600
A821	43041	Uint16	GC_AF_PF_ABS_MIN	Advance Function PF Min Set (Abs Value) PFmax = 1, Pflmin = PFlending or PFlagging(Abs)	0.1%	RW	800	1000	800
A822	43042	Uint16	Reserved	/	/	/	/	/	/
A823	43043	Uint16	GC_AF_QMODE_ENTRY_P_ABS	Advance Function Q Mode Entrance Condition of P Function For: SPF/VV(Q)	0.1%	RW	0	1000	50
A824	43044	Uint16	GC_AF_QMODE_EXIT_P_ABS	Advance Function Q Mode Exit Condition of P Function For: SPF/VV(Q)	0.1%	RW	0	1000	40
A825	43045	Uint16	GC_SPF_MODE_EN	Specified Power Factor Mode. 0: Disabled. 1: Enable with PF Priority..	NA	RW	0	1	0
A826	43046	Int16	GC_SPF_TARGET	Displacement Power Factor Target Value.	0.1%	RW	-1000	1000	1000
A827	43047	Uint16	GC_SPF_EXCITATION	System excited state 0: System is over-excited 1: System is under-excited	NA	RW	0	1	0
A828	43048	Uint16	GC_SPF_RESPON_TIME	PF Response Time Setting. 0: Immediately. Other: Unit, 0.01s	0.01s	RW	0	10000	1000

Note: The power direction is defined as follows:

P+: active power flows from PCS to grid.

P-: active power flows from grid to PCS

Q+: Inductive reactive power flows from PCS to grid (over-excited)

Q-: Capacitive reactive power flows from PCS to grid (under-excited)

4.3.7. Volt-VAR

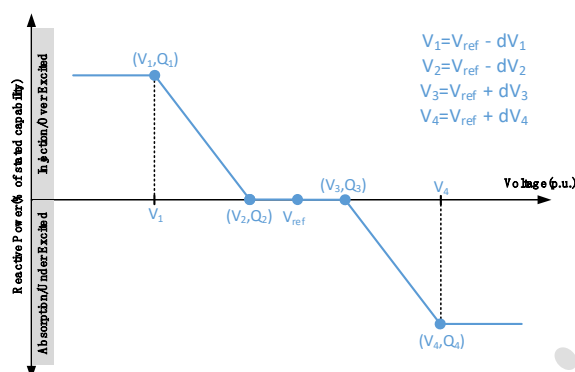


Figure 4-4 Volt-Var Control Mode Curve

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A833	43059	UInt16	GC_VVAR_MODE_EN	Voltage Vs Var Function Switch. 0: Disable. 1: Enable with 4 Points Mode.	NA	RW	0	1	0
A834	43060	UInt16	GC_VVAR_V1	VV Function dV1	0.1%	RW	20	180	80
A835	43061	Int16	GC_VVAR_Q1	VV Function Q1	0.1%	RW	0	600	440
A836	43062	UInt16	GC_VVAR_V2	VV Function dV2	0.1%	RW	0	30	20
A837	43063	Int16	GC_VVAR_Q2	VV Function Q2	0.1%	RW	-600	600	0
A838	43064	UInt16	GC_VVAR_V3	VV Function dV3	0.1%	RW	0	30	20
A839	43065	Int16	GC_VVAR_Q3	VV Function Q3	0.1%	RW	-600	600	0
A83A	43066	UInt16	GC_VVAR_V4	VV Function dV4	0.1%	RW	20	180	80
A83B	43067	Int16	GC_VVAR_Q4	VV Function Q4	0.1%	RW	-600	0	-440
A83C- A839	43068- 43071	UInt16	Reserved	NA	NA	NA	NA	NA	NA
A840	43072	UInt16	GC_VVAR_RESPON_TIME	Volt-Var Response Time setting. 0: Immediately. Other: Unit, 0.01s	0.01s	RW	100	9000	500
A841	43073	UInt16	GC_VVAR_V_FILTER_TIME	Volt-Var V _{ref} Filter Time setting. 0: Immediately. Other: Unit, s	s	RW	1	65535	300
A842	43074	UInt16	GC_VVAR_VREF_ADJUSTMEN T_EN	Volt-Var V _{ref} autonomous adjustment. 0: Disable, V _{ref} = value of reg 43073 1: Enable, V _{ref} is the grid value after filtering based on the time constant (reg 41069).	NA	RW	0	1	0
A843	43075	UInt16	GC_VVAR_VREF	Volt-Var V _{ref} setting	0.1%	RW	950	1050	1000

4.3.8. Voltage-Watt

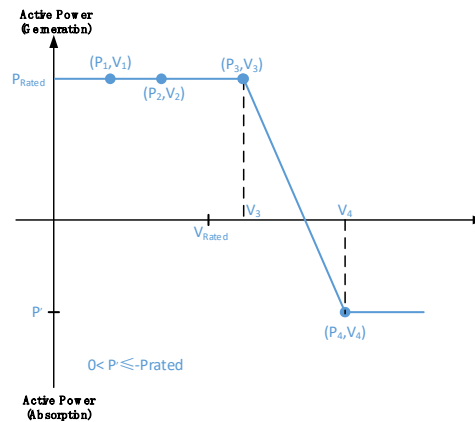


Figure 4-5 Volt-Watt Control Mode Limit Area

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A84A	43082	UInt16	GC_VWATT_MODE_EN	Voltage Vs Watt Function Switch. 0: Disable. 1: Enable	NA	RW	0	1	0
A84B	43083	UInt16	GC_VWATT_V1	VW Function Voltage Point 1	0.1%	RW	900	1000	900
A84C	43084	Int16	GC_VWATT_P1	VW Function Active Power Point 1	0.1%	RW	1000	1000	1000
A84D	43085	UInt16	GC_VWATT_V2	VW Function Voltage Point 2 Under Voltage Start Point	0.1%	RW	900	1000	950
A84E	43086	Int16	GC_VWATT_P2	VW Function Active Power Point 2	0.1%	RW	1000	1000	1000
A84F	43087	UInt16	GC_VWATT_V3	VW Function Voltage Point 3 Over Voltage Start Point	0.1%	RW	1050	1090	1060
A850	43088	Int16	GC_VWATT_P3	VW Function Active Power Point 3	0.1%	RW	1000	1000	1000
A851	43089	UInt16	GC_VWATT_V4	VW Function Voltage Point 4	0.1%	RW	1060	1100	1100
A852	43090	Int16	GC_VWATT_P4	VW Function Active Power Point 4	0.1%	RW	-1000	0	-1000
A853	43091	UInt16	GC_VWATT_RESPON_TIME	VW Response Time Setting. 0: Immediately. Other: Unit, 0.01s	0.01s	RW	50	6000	1000

4.3.9. Frequency-Watt

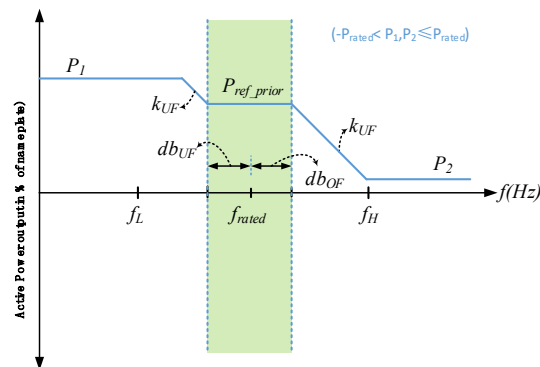


Figure 4-6 Freq-Watt Control Mode Curve

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A85A	43098	Uint16	GC_FWATT_OF_MODE_EN	Freq-watt Function Switch for Over-frequency. 0: Disable. 1: Enable	NA	RW	0	1	0
A85B	43099	Uint16	GC_FWATT_OF_DEADBAND	Frequency Droop Dead-band for Over-frequency Conditions (db_{OF})	0.1%	RW	0	1000	36
A85C	43100	Uint16	GC_FWATT_UF_DEADBAND	Frequency Droop Dead-band for Under-frequency Conditions (db_{UF})	0.1%	RW	0	1000	36
A85D	43101	Uint16	GC_FWATT_OF_DROOP_RATIO	Frequency Droop per-unit Change ratio for Over-Frequency Conditions (k_{OF}) Note: Corresponding to 1 per-unit Power Output Change.	0.1%	RW	0	100	50
A85E	43102	Uint16	GC_FWATT_UF_DROOP_RATIO	Frequency Droop per-unit Change Ratio for Under-Frequency Conditions. (k_{UF}) Note: Corresponding to 1 per-unit Power Output Change	0.1%	RW	0	100	50
A85F	43103	Uint16	GC_FWATT_RESPON_TIME	Freq-Watt Response Time Setting. 0: Immediately. Other: Unit, 0.01s	0.01s	RW	0	1000	500
A860- A864	43104- 41108	Uint16	Reserved	/	/	/	/	/	/
A865	43109	Uint16	GC_FWATT_UF_MODE_EN	Freq-watt Function Switch for Under-frequency. 0: Disable. 1: Enable	NA	RW	0	1	0

4.3.10. High/Low Voltage Ride Through

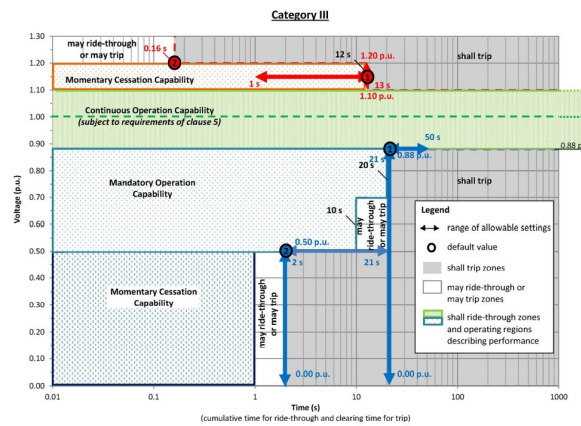


Figure 4-7 Voltage Ride-through operating performance(Category III)

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A87B	43131	Uint16	GC_AC_OVRT_MODE_EN	Over Voltage Ride Through Mode Selection/Enable 0: Disable. 1: Enable with power limitation.	NA	RW	0	1	0
A87C	43132	Uint16	GC_AC_UVRT_MODE_EN	Under Voltage Ride Through Mode Selection/Enable. 0: Disable. 1: Enable with power limitation.	NA	RW	0	1	0
A87D	43133	Uint16	GC_AC_OVRT_HV2	OVRT Level 2 Voltage Setting	0.1%	RW	1100	1300	1200
A87E	43134	Uint16	GC_AC_OVRT_HV1	OVRT Level 1 Voltage Setting	0.1%	RW	1000	1100	1100
A87F	43135	Uint16	GC_AC_UVRT_LV1	UVRT Level 1 Voltage Setting	0.1%	RW	700	1000	880
A880	43136	Uint16	GC_AC_UVRT_LV2	UVRT Level 2 Voltage Setting	0.1%	RW	0	880	500
A881	43137	Uint16	GC_AC_UVRT_LV3	UVRT Level 3 Voltage Setting	0.1%	RW	0	500	500

4.3.11. High/Low Frequency Ride Through

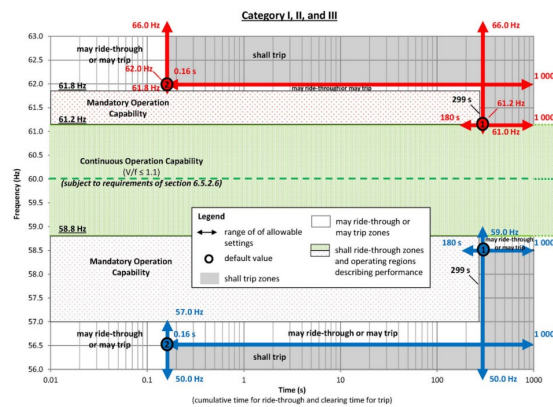


Figure 4-8 Frequency ride-through operating performance

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A888	43144	UInt16	GC_AC_OUFRT_EN	Over/Under Frequency Ride Through Mode Selection/Enable 0: Disable. 1: Enable.	NA	RW	0	1	0
A889	43145	UInt16	GC_AC_OFRT_HF2	OFRT Level 2 Frequency Setting	0.01Hz	RW	6120	6200	6180
A88A	43146	UInt16	GC_AC_OFRT_HF1	OFRT Level 1 Frequency Setting	0.01Hz	RW	6000	6120	6120
A88B	43147	UInt16	GC_AC_UFRT_LF1	UFRT Level 1 Frequency Setting	0.01Hz	RW	5880	6000	5880
A88C	43148	UInt16	GC_AC_UFRT_LF2	UFRT Level 2 Frequency Setting	0.01Hz	RW	5700	5880	5700

4.3.12. Watt-Var

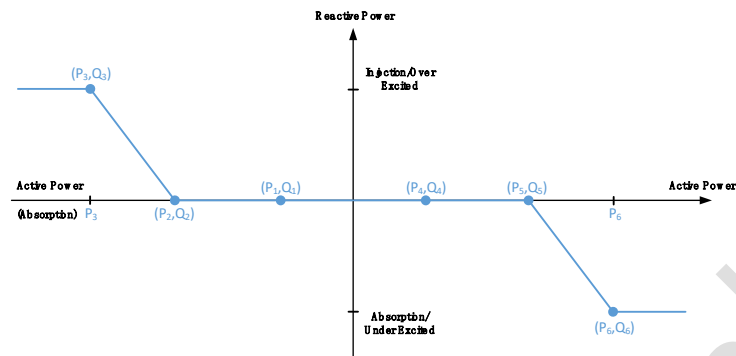


Figure 4-9 Active power-reactive power characteristic

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A891	43153	Uint16	GC_WATTVAR_MODE_EN	Enable Active Power-Reactive Power Mode. 0: Disable. 1: Enable.	NA	RW	0	1	0
A892	43154	Int16	GC_WATTVAR_P1	WVAR Function P1	0.1%	RW	-1000	-500	-200
A893	43155	Int16	GC_WATTVAR_Q1	WVAR Function Q1	0.1%	RW	-600	600	440
A894	43156	Int16	GC_WATTVAR_P2	WVAR Function P2	0.1%	RW	-800	-400	-500
A895	43157	Int16	GC_WATTVAR_Q2	WVAR Function Q2	0.1%	RW	-600	600	0
A896	43158	Int16	GC_WATTVAR_P3	WVAR Function P3	0.1%	RW	-1000	-300	-200
A897	43159	Int16	GC_WATTVAR_Q3	WVAR Function Q3	0.1%	RW	-600	600	0
A898	43160	Int16	GC_WATTVAR_P4	WVAR Function P4	0.1%	RW	0	700	200
A899	43161	Int16	GC_WATTVAR_Q4	WVAR Function Q4	0.1%	RW	-600	600	0
A89A	43162	Int16	GC_WATTVAR_P5	WVAR Function P5	0.1%	RW	400	800	500
A89B	43163	Int16	GC_WATTVAR_Q5	WVAR Function Q5	0.1%	RW	-600	600	0
A89C	43164	Int16	GC_WATTVAR_P6	WVAR Function P6	0.1%	RW	500	1000	1000
A89D	43165	Int16	GC_WATTVAR_Q6	WVAR Function Q6	0.1%	RW	-600	600	-440
A89E	43166	Uint16	GC_WATTVAR_RESPON_TIME	WVAR Response Time Setting.	0.01s	RW	0	1000	500

4.3.13. Limit Active Power

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A89F	43167	Uint16	GC_LIMIT_ACTIVE_POWER_EN	Limit active power mode Enable 0: Disable. 1: Enable .	NA	RW	0	1	0
A8A0	43168	Uint16	GC_LIMIT_ACTIVE_POWER_VAULT	Active Power Limitation Unit: 0.1%	0.1%	RW	0	1100	1000

Delta Confidential

4.3.14. Permit Service

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A8A2	43170	Uint16	GC_ENTER_SERVICE_EN	Enter Service Enable. 0: Disable. 1: Enable .	NA	RW	0	1	0

Delta Confidential

4.3.15. Anti-Islanding Protection

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A8A3	43171	UInt16	GC_ANTI_ISLANDING_EN	Anti-islanding Check Enable bit 0: Active method bit 1: Passive method =0: Disable. =1: Enable.	NA	RW	0	1	0

5. Operation Command

Note: Only need to send command to the center controller, that is the center controller of string system, and then the center controller will copy it to each power module through internal high speed fiber bus.

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default	OA																																																																																																
A5A0	42400	Uint16	CMD_SYS_ON_OFF	System power on/off command 0: OFF 1: ON <i>Note: Valid only if Reg. 42207 is set to 0.</i>	NA	RW	0	2	0	Yes																																																																																																
A5A1	42401	Uint16	CMD_UNIT_ON_OFF	Power unit on/off command bitx=0: OFF bitx=1: ON x=1..12 <i>Note: Valid only if Reg. 42207 is set to 1.</i>	NA	RW	0	4095	0	Yes																																																																																																
<table><tr><th colspan="2">bit 15</th><th colspan="2">bit 14</th><th colspan="2">bit 13</th><th colspan="2">bit 12</th><th colspan="2">bit 11</th><th colspan="2">bit 10</th><th colspan="2">bit 9</th><th colspan="2">bit 8</th></tr><tr><th colspan="2">NA</th><th colspan="2">NA</th><th colspan="2">NA</th><th colspan="2">NA</th><th colspan="2">Unit #12</th><th colspan="2">Unit #11</th><th colspan="2">Unit #10</th><th colspan="2">Unit #9</th></tr><tr><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td></tr><tr><th colspan="2">bit 7</th><th colspan="2">bit 6</th><th colspan="2">bit 5</th><th colspan="2">bit 4</th><th colspan="2">bit 3</th><th colspan="2">bit 2</th><th colspan="2">bit 1</th><th colspan="2">bit 0</th></tr><tr><td colspan="2">Unit #8</td><td colspan="2">Unit #7</td><td colspan="2">Unit #6</td><td colspan="2">Unit #5</td><td colspan="2">Unit #4</td><td colspan="2">Unit #3</td><td colspan="2">Unit #2</td><td colspan="2">Unit #1</td></tr><tr><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td><td colspan="2">RW-0h</td></tr></table>											bit 15		bit 14		bit 13		bit 12		bit 11		bit 10		bit 9		bit 8		NA		NA		NA		NA		Unit #12		Unit #11		Unit #10		Unit #9		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		bit 7		bit 6		bit 5		bit 4		bit 3		bit 2		bit 1		bit 0		Unit #8		Unit #7		Unit #6		Unit #5		Unit #4		Unit #3		Unit #2		Unit #1		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h	
bit 15		bit 14		bit 13		bit 12		bit 11		bit 10		bit 9		bit 8																																																																																												
NA		NA		NA		NA		Unit #12		Unit #11		Unit #10		Unit #9																																																																																												
RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h																																																																																												
bit 7		bit 6		bit 5		bit 4		bit 3		bit 2		bit 1		bit 0																																																																																												
Unit #8		Unit #7		Unit #6		Unit #5		Unit #4		Unit #3		Unit #2		Unit #1																																																																																												
RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h		RW-0h																																																																																												
A5A2	42402	Int16	Reserved	NA	/	/	/	/	/	/																																																																																																
A5A3	42403	Int16	CMD_SYS_P_DEMAND	System Power Demand: Active Power. Unit: 1kw <i>Note: Valid only if Reg. 42207 is set to 0.</i>	1kW	R/W	-3000	3000	0	Yes																																																																																																
A5A4	42404	Int16	CMD_SYS_Q_DEMAND	System Power Demand: Reactive Power. Unit: 1kvar <i>Note: Valid only if Reg. 42207 is set to 0.</i>	1kVar	R/W	-3000	3000	0	Yes																																																																																																
A5A5	42405	Int16	CMD_UNIT1_P_DEMAND	Power Demand: Active Power Of Unit #1. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes																																																																																																
A5A6	42406	Int16	CMD_UNIT1_Q_DEMAND	Power Demand: Reactive Power Of Unit #1. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes																																																																																																
A5A7	42407	Int16	CMD_UNIT2_P_DEMAND	Power Demand: Active Power Of Unit #2. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes																																																																																																
A5A8	42408	Int16	CMD_UNIT2_Q_DEMAND	Power Demand: Reactive Power Of Unit #2. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes																																																																																																
A5A9	42409	Int16	CMD_UNIT3_P_DEMAND	Power Demand: Active Power Of Unit #3. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes																																																																																																
A5AA	42410	Int16	CMD_UNIT3_Q_DEMAND	Power Demand: Reactive Power Of Unit #3. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes																																																																																																
A5AB	42411	Int16	CMD_UNIT4_P_DEMAND	Power Demand: Active Power Of Unit #4. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes																																																																																																
A5AC	42412	Int16	CMD_UNIT4_Q_DEMAND	Power Demand: Reactive Power Of Unit #4. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes																																																																																																
A5AD	42413	Int16	CMD_UNIT5_P_DEMAND	Power Demand: Active Power Of Unit #5. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes																																																																																																
A5AE	42414	Int16	CMD_UNIT5_Q_DEMAND	Power Demand: Reactive Power Of Unit #5.	1kVar	R/W	-3000	3000	0	Yes																																																																																																

				Unit: 1kvar						
A5AF	42415	Int16	CMD_UNIT6_P_DEMAND	Power Demand: Active Power Of Unit #6. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5B0	42416	Int16	CMD_UNIT6_Q_DEMAND	Power Demand: Reactive Power Of Unit #6. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5B1	42417	Int16	CMD_UNIT7_P_DEMAND	Power Demand: Active Power Of Unit #7. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5B2	42418	Int16	CMD_UNIT7_Q_DEMAND	Power Demand: Reactive Power Of Unit #7. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5B3	42419	Int16	CMD_UNIT8_P_DEMAND	Power Demand: Active Power Of Unit #8. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5B4	42420	Int16	CMD_UNIT8_Q_DEMAND	Power Demand: Reactive Power Of Unit #8. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5B5	42421	Int16	CMD_UNIT9_P_DEMAND	Power Demand: Active Power Of Unit #9. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5B6	42422	Int16	CMD_UNIT9_Q_DEMAND	Power Demand: Reactive Power Of Unit #9. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5B7	42423	Int16	CMD_UNIT10_P_DEMAND	Power Demand: Active Power Of Unit #10. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5B8	42424	Int16	CMD_UNIT10_Q_DEMAND	Power Demand: Reactive Power Of Unit #10. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5B9	42425	Int16	CMD_UNIT11_P_DEMAND	Power Demand: Active Power Of Unit #11. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5BA	42426	Int16	CMD_UNIT11_Q_DEMAND	Power Demand: Reactive Power Of Unit #11. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5BB	42427	Int16	CMD_UNIT12_P_DEMAND	Power Demand: Active Power Of Unit #12. Unit: 1kw	1kW	R/W	-3000	3000	0	Yes
A5BC	42428	Int16	CMD_UNIT12_Q_DEMAND	Power Demand: Reactive Power Of Unit #12. Unit: 1kvar	1kVar	R/W	-3000	3000	0	Yes
A5D5	42453	Uint16	CMD_SYS_FAULT_RESET	Clear all faults of string system and unit/module and/or reset controller. Bit0=1: Reset all faults Bit1=1: Hardware reset, Reset and restart all DSP controllers, including system controller and module controller of sub-system Others: Reserved	NA	R/W	0	65535	0	No

Note: Reg. 42405 to 42428 are valid when Reg. 42207 is set to 1.

6. System Operation Status

6.1. Running Status Info.

6.1.1. Center Controller

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default																																																
7596	30102	Uint16	OS_COMM_STATE	Site controller inner communication state, bits definition, see table: 0: Disable 1: Normal 2: Fault	NA	R	0	65535	0																																																
<table><tr><td>bit 15</td><td>bit 14</td><td>bit 13</td><td>bit 12</td><td>bit 11</td><td>bit 10</td><td>bit 9</td><td>bit 8</td></tr><tr><td colspan="2">NA</td><td colspan="2">ETH1 port</td><td colspan="2">ETH0 port</td><td colspan="2">RS485C-reserved</td></tr><tr><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td></tr><tr><td>bit 7</td><td>bit 6</td><td>bit 5</td><td>bit 4</td><td>bit 3</td><td>bit 2</td><td>bit 1</td><td>bit0</td></tr><tr><td colspan="2">RS485B-reserved</td><td colspan="2">RS485A-reserved</td><td colspan="2">CAN-reserved</td><td colspan="2">SPI</td></tr><tr><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td></tr></table>										bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8	NA		ETH1 port		ETH0 port		RS485C-reserved		R-0h		R-0h		R-0h		R-0h		bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0	RS485B-reserved		RS485A-reserved		CAN-reserved		SPI		R-0h		R-0h		R-0h		R-0h	
bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8																																																		
NA		ETH1 port		ETH0 port		RS485C-reserved																																																			
R-0h		R-0h		R-0h		R-0h																																																			
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0																																																		
RS485B-reserved		RS485A-reserved		CAN-reserved		SPI																																																			
R-0h		R-0h		R-0h		R-0h																																																			
7597	30103	Uint16	OS_SYS_ETH_STATE1	Intranet communication state, it mainly refers to the Ethernet communication status between center controller, system controller and power modules. bits definition, see table 0: Disable 1: Normal 2: Fault	NA	R	0	65535	0																																																
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bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8																																																		
Power module #7 controller Ethernet state		Power module #6 controller Ethernet state		Power module #5 controller Ethernet state		Power module #4 controller Ethernet state																																																			
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bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0																																																		
Power module #3 controller Ethernet state		Power module #2 controller Ethernet state		Power module #1 controller Ethernet state		System controller Ethernet state																																																			
R-0h		R-0h		R-0h		R-0h																																																			
7598	30104	Uint16	OS_SYS_ETH_STATE2	Intranet communication state, it mainly refers to the Ethernet communication status between center controller, system controller and power modules. bits definition, see table below 0: Disable 1: Normal 2: Fault	NA	R	0	65535	0																																																
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R-0h		R-0h		R-0h		R-0h																																																			
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0																																																		
Power module #11 controller Ethernet state		Power module #10 controller Ethernet state		Power module #9 controller Ethernet state		Power module #8 controller Ethernet state																																																			
R-0h		R-0h		R-0h		R-0h																																																			

6.1.2. System & Module Controller

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
75F8	30200	Unit16	SYS_GLB_STATE	System Global State bit0: Initial bit1: Standby bit2: Starting bit3: Normal bit4: Stopping bit5: Fault	NA	R	0	65535	0
75F9	30201	Unit16	Reserved	/	/	/	/	/	/
75FA	30202	Unit16	SYS_IO_STATE	Sub-System I/O State Bit0: Low Voltage Switcher State Bit1: High Voltage Switcher State Bit2: Load Branch Switcher State Bit3: System Disable Bit4: FAN Contactor 1 State Bit5: FAN Contactor 2 State Bit6: Humidity Switcher State Bit7: EPO State Bit8: Remote EPO Fault Bit9: IMD State Bit10: 220Vac Power State Bit11: CM Power State Bit12: IO Power State Bit13: Ctrl Power State Bit14: 24V Power State Bit15: Aux Transformer State	NA	R	0	65535	0
75FB	30203	Unit16	POW_UNIT#1_OPT_STATE	Power Unit #1 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
75FC	30204	Unit16	POW_UNIT#2_OPT_STATE	Power Unit #2 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
75FD	30205	Unit16	POW_UNIT#3_OPT_STATE	Power Unit #3 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
75FE	30206	Unit16	POW_UNIT#4_OPT_STATE	Power Unit #4 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
75FF	30207	Unit16	POW_UNIT#5_OPT_STATE	Power Unit #5 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7600	30208	Unit16	POW_UNIT#6_OPT_STATE	Power Unit #6 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7601	30209	Unit16	POW_UNIT#7_OPT_STATE	Power Unit #7 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0

7602	30210	Uint16	POW_UNIT#8_OPT_STATE	Power Unit #8 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7603	30211	Uint16	POW_UNIT#9_OPT_STATE	Power Unit #9 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7604	30212	Uint16	POW_UNIT#10_OPT_STATE	Power Unit #10 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7605	30213	Uint16	POW_UNIT#11_OPT_STATE	Power Unit #11 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7606	30214	Uint16	POW_UNIT#12_OPT_STATE	Power Unit #12 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7607	30215	Uint16	POW_MOD#1_OPT_STATE	Power Module #1 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7608	30216	Uint16	POW_MOD#2_OPT_STATE	Power Module #2 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
7609	30217	Uint16	POW_MOD#3_OPT_STATE	Power Module #3 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
760A	30218	Uint16	POW_MOD#4_OPT_STATE	Power Module #4 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
760B	30219	Uint16	POW_MOD#5_OPT_STATE	Power Module #5 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
760C	30220	Uint16	POW_MOD#6_OPT_STATE	Power Module #6 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
760D	30221	Uint16	POW_MOD#7_OPT_STATE	Power Module #7 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0
760E	30222	Uint16	POW_MOD#8_OPT_STATE	Power Module #8 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0

760F	30223	Uint16	POW_MOD#9_OPT_STATE	Power Module #9 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0																																																
7610	30224	Uint16	POW_MOD#10_OPT_STATE	Power Module #10 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0																																																
7611	30225	Uint16	POW_MOD#11_OPT_STATE	Power Module #11 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0																																																
7612	30226	Uint16	POW_MOD#12_OPT_STATE	Power Module #12 Operation State 0: Initial 1: Standby 2: Starting 3: Normal 4: Fault	NA	R	0	16	0																																																
7613	30227	Uint16	SYS_FIBER_COMM_STATE1	Inner fiber communication state, it mainly refers to the high-speed communication status between sub-system controller and power modules controller. bits definition, see table 0: Disable 1: Normal 2: Fault	NA	R	0	65535	0																																																
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bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8																																																		
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R-0h		R-0h		R-0h		R-0h																																																			
7614	30228	Uint16	SYS_FIBER_COMM_STATE2	Inner fiber communication state, it mainly refers to the high-speed communication status between sub-system controller and power modules controller. bits definition, see table 0: Disable 1: Normal 2: Fault	NA	R	0	65535	0																																																
<table><tr><td>bit 15</td><td>bit 14</td><td>bit 13</td><td>bit 12</td><td>bit 11</td><td>bit 10</td><td>bit 9</td><td>bit 8</td></tr><tr><td colspan="2">NA</td><td colspan="2">NA</td><td colspan="2">NA</td><td colspan="2">NA</td></tr><tr><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td></tr><tr><td>bit 7</td><td>bit 6</td><td>bit 5</td><td>bit 4</td><td>bit 3</td><td>bit 2</td><td>bit 1</td><td>bit0</td></tr><tr><td colspan="2">Power Module #12 Fiber Comm. State</td><td colspan="2">Power Module #10 Fiber Comm. State</td><td colspan="2">Power Module #9 Fiber Comm. State</td><td colspan="2">Power Module #8 Fiber Comm. State</td></tr><tr><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td><td colspan="2">R-0h</td></tr></table>										bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8	NA		NA		NA		NA		R-0h		R-0h		R-0h		R-0h		bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0	Power Module #12 Fiber Comm. State		Power Module #10 Fiber Comm. State		Power Module #9 Fiber Comm. State		Power Module #8 Fiber Comm. State		R-0h		R-0h		R-0h		R-0h	
bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9	bit 8																																																		
NA		NA		NA		NA																																																			
R-0h		R-0h		R-0h		R-0h																																																			
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit0																																																		
Power Module #12 Fiber Comm. State		Power Module #10 Fiber Comm. State		Power Module #9 Fiber Comm. State		Power Module #8 Fiber Comm. State																																																			
R-0h		R-0h		R-0h		R-0h																																																			

6.2. Key Variable Monitoring

6.2.1. System Monitoring

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
759F	30111	Uint16	OS_MEAS_IMD_ISO	Sting PCS System Insulation resistance	kOhm	R	0	1000	0
766C	30316	Uint16	SYS_MEAS_GRID_VRS	System Vgrid RS	0.1V	R	0	10000	0
766D	30317	Uint16	SYS_MEAS_GRID_VST	System Vgrid ST	0.1V	R	0	10000	0
766E	30318	Uint16	SYS_MEAS_GRID_VTR	System Vgrid TR	0.1V	R	0	10000	0
766F	30319	Uint16	SYS_MEAS_GRID_FREQ	System Frequency	0.01Hz	R	0	7000	0
7670	30320	Uint16	SYS_MEAS_IO_R	System Output Ir	0.1A	R	0	60000	0
7671	30321	Uint16	SYS_MEAS_IO_S	System Output Is	0.1A	R	0	60000	0
7672	30322	Uint16	SYS_MEAS_IO_T	System Output It	0.1A	R	0	60000	0
7673	30323	Int16	SYS_MEAS_P	System active power	1kW	R	-32767	32767	0
7674	30324	Int16	SYS_MEAS_Q	System Reactive Power	1kVar	R	-32767	32767	0
7675	30325	Uint16	SYS_MEAS_S	System Apparent Power	1kVA	R	0	65535	0
7676	30326	Int16	SYS_AVAILABLE_P	System Available Active Power	1kW	R	-32767	32767	0
7677	30327	Int16	Reserved	/	/	/	/	/	/
7678	30328	Uint16	SYS_AVAILABLE_S	System Available Apparent Power	1kVA	R	0	60000	0
7679	30329	Int16	SYS_MEAS_PF	System Power Factor	0.001	R	-1000	1000	0

6.2.2. Power Units Monitoring

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
Power Unit #1 Monitoring (start address:30354, length: 4 words)									
7696	30354	Int16	POW_UNIT1_MEAS_P	Unit #1 Active Power	0.1kW	R	-30000	30000	0
7697	30355	Int16	POW_UNIT1_MEAS_Q	Unit #1 Reactive Power	0.1kVar	R	-30000	30000	0
7698	30356	Int16	POW_UNIT1_AVAILABLE_P	Unit #1 Available Active Power	0.1kW	R	0	60000	0
7699	30357	Uint16	POW_UNIT1_AVAILABLE_S	Unit #1 Available Apparent Power	0.1kVA	R	0	60000	0
Power Unit #n Monitoring (start address:30354+(n-1)*4, n=2 to 12, length:4 words)									
/	/	Int16	POW_UNIT#n_MEAS_P	Unit #n Active Power	0.1kW	R	-30000	30000	0
/	/	Int16	POW_UNIT#n_MEAS_Q	Unit #n Reactive Power	0.1kVar	R	-30000	30000	0
/	/	Int16	POW_UNIT#n_AVAILABLE_P	Unit #n Available Active Power	0.1kW	R	0	60000	0
/	/	Uint16	POW_UNIT#n_AVAILABLE_S	Unit #n Available Apparent Power	0.1kVA	R	0	60000	0

6.2.3. Power Modules Monitoring

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
Power Module #1 Start Address: 31010 Length: 33									
7986	31010	Uint16	POW_MOD1_MEAS_GRID_VRS	Power module voltage RS	0.1V	R	0	9000	0
7987	31011	Uint16	POW_MOD1_MEAS_GRID_VST	Power module voltage ST	0.1V	R	0	9000	0
7988	31012	Uint16	POW_MOD1_MEAS_GRID_VTR	Power module voltage TR	0.1V	R	0	9000	0
7989	31013	Uint16	POW_MOD1_MEAS_CAP_VRS	Power module cap voltage RS	0.1V	R	0	9000	0
798A	31014	Uint16	POW_MOD1_MEAS_CAP_VST	Power module cap voltage ST	0.1V	R	0	9000	0
798B	31015	Uint16	POW_MOD1_MEAS_CAP_VTR	Power module cap voltage TR	0.1V	R	0	9000	0
798C	31016	Uint16	POW_MOD1_MEAS_GRID_FRE Q	Power module Frequency	0.01Hz	R	0	7000	0
798D	31017	Uint16	POW_MOD1A_MEAS_IO_R	INV1 Output Current Ir	0.1A	R	0	4000	0
798E	31018	Uint16	POW_MOD1A_MEAS_IO_S	INV1 Output Current Is	0.1A	R	0	4000	0
798F	31019	Uint16	POW_MOD1A_MEAS_IO_T	INV1 Output Current It	0.1A	R	0	4000	0
7990	31020	Uint16	POW_MOD1B_MEAS_IO_R	INV2 Output Current Ir	0.1A	R	0	4000	0
7991	31021	Uint16	POW_MOD1B_MEAS_IO_S	INV2 Output Current Is	0.1A	R	0	4000	0
7992	31022	Uint16	POW_MOD1B_MEAS_IO_T	INV2 Output Current It	0.1A	R	0	4000	0
7993	31023	Uint16	POW_MOD1_MEAS_IO_R	INV2 Output Current Ir	0.1A	R	0	4000	0
7994	31024	Uint16	POW_MOD1_MEAS_IO_S	Output Current Is	0.1A	R	0	4000	0
7995	31025	Uint16	POW_MOD1_MEAS_IO_T	Output Current It	0.1A	R	0	4000	0
7996	31026	Int16	POW_MOD1_MEAS_P	Module AC Active Power	0.1kW	R	-3000	3000	0
7997	31027	Int16	POW_MOD1_MEAS_Q	Module AC Reactive Power	0.1kVar	R	-3000	3000	0
7998	31028	Uint16	POW_MOD1_MEAS_S	Module AC Apparent Power	0.1kVA	R	0	3000	0
7999	31029	Int16	POW_MOD1_MEAS_PF	Module Power Factor Note: when apparent power is less than 1kVA, this value will be set to 1001, which means the value is invalid.	NA	R	-1000	1000	0
799A	31030	Int16	POW_MOD1_MEAS_DC_VIN	Module DC Input Voltage	0.1V	R	-20000	20000	0
799B	31031	Int16	POW_MOD1_MEAS_DC_IIN	Module DC Input Current	0.1A	R	-4000	4000	0
799C	31032	Int16	POW_MOD1_MEAS_DC_BUS	Module DC Bus Voltage	0.1V	R	0	20000	0
799D	31033	Int16	POW_MOD1_MEAS_T_ABD	A Board Temperature	0.1°C	R	-1000	2000	0

799E	31034	Uint16	POW_MOD1_MEAS_V_PCAP	Power Module XCap Phase Voltage	0.1V	R	0	7000	0
799F	31035	Int16	POW_MOD1_MEAS_T_CBD	C Board Temperature	0.1°C	R	-1000	2000	0
79A0	31036	Int16	POW_MOD1_MEAS_T_INLET	Inlet Temperature	0.1°C	R	-1000	2000	0
79A1	31037	Int16	POW_MOD1A_MEAS_TINV_R	INV1 bridge R temperature	0.1°C	R	-1000	2000	0
79A2	31038	Int16	POW_MOD1A_MEAS_TINV_S	INV1 bridge S temperature	0.1°C	R	-1000	2000	0
79A3	31039	Int16	POW_MOD1A_MEAS_TINV_T	INV1 bridge T temperature	0.1°C	R	-1000	2000	0
79A4	31040	Int16	POW_MOD1B_MEAS_TINV_R	INV2 bridge R temperature	0.1°C	R	-1000	2000	0
79A5	31041	Int16	POW_MOD1B_MEAS_TINV_S	INV2 bridge S temperature	0.1°C	R	-1000	2000	0
79A6	31042	Int16	POW_MOD1B_MEAS_TINV_T	INV2 bridge T temperature	0.1°C	R	-1000	2000	0
Power Module #2 Start Address: 31110 Length: 33									
Power Module #3 Start Address: 31210 Length: 33									
Power Module #4 Start Address: 31310 Length: 33									
Power Module #5 Start Address: 31410 Length: 33									
Power Module #6 Start Address: 31510 Length: 33									
Power Module #7 Start Address: 31610 Length: 33									
Power Module #8 Start Address: 31710 Length: 33									
Power Module #9 Start Address: 31810 Length: 33									
Power Module #10 Start Address: 31910 Length: 33									
Power Module #11 Start Address: 32010 Length: 33									
Power Module #12 Start Address: 32110 Length: 33									

7. Alarm & Warning

7.1. System Alarm

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	Bit	DESCRIPTION	Action	RW	Min	Max	Default
765E	30302	Uint16	SYS_FLT_HARDWARE_1	0	LV switch feedback fault	PCS Off	R	0	65535	0
				1	HV grid switch feedback fault					
				2	HV load switch feedback fault					
				3	EPO fault					
				4	External fault input					
				5	Sys IMD fault					
				6	230V power failure fault					
				7	I/O power failure fault					
				8	Control power failure fault					
				9	Communication power failure fault					
				10	24V power failure fault					
				11	Y board comm fault					
				12	IMD RS485 Loss fault					
				13	DSP EMIF Write fault					
				14	DC Combiner OTP fault					
				15	Reserved	NA				
765F	30303	Uint16	SYS_FLT_HARDWARE_2	0	Group B DSP SPI Comm fault	PCS Off	R	0	65535	0
				1	Fan Contactor1 feedback fault					
				2	Fan Contactor2 feedback fault					
				3	Group All Module fault					
				4	Center controller SPI Comm fault					
				5	PMTB fault					
				6	PMTA fault					
				7	Group B DSP by Sys Off fault					

				8	DC1_Cabinet Amb OTP fault					
				9	DC2_Cabinet Amb OTP fault					
				10	Reserved	NA				
				11	Reserved	NA				
				12	Reserved	NA				
				13	Reserved	NA				
				14	Reserved	NA				
				15	Reserved	NA				
7660	30304	Uint16	SYS_FLT_GRID_1	0	Grid Voltage OVP	PCS Off	R	0	65535	0
				1	Grid Voltage UVP					
				2	Grid Voltage OFP					
				3	Grid Voltage UFP					
				4	Phase sequence Error					
				5	Grid Voltage Transient OVP					
				6	Load Current RMS OCP					
				7	Load Current Transient OCP					
				8	Load Current Zero-seq OCP					
				9	Load Current Unbalance					
				10	Anti-Islanding fault					
				11	Voltage DC offset fault					
				12	Current DC offset fault					
				13	Reserved	NA				
				14	Reserved	NA				
				15	Reserved	NA				
7661	30305	Uint16	SYS_FLT_BAT1	0	BMS Comm loss fault	PCS Off	R	0	65535	0
				1	DC Input 1 Voltage UVP					
				2	DC Input 2 Voltage OVP					
				3	DC Input 2 Voltage UVP					
				4	DC Input 3 Voltage OVP					
				5	DC Input 3 Voltage UVP					
				6	DC Input 4 Voltage OVP					
				7	DC Input 4 Voltage UVP					
				8	DC Input 5 Voltage OVP					
				9	DC Input 5 Voltage UVP					
				10	DC Input 6 Voltage OVP					
				11	DC Input 6 Voltage UVP					
				12	DC Input 1 Current OCP					

				13	DC Input 2 Current OCP					
				14	DC Input 3 Current OCP					
				15	DC Input 4 Current OCP					
7662	30306	Uint16	SYS_FLT_YBOARD_1	0	Trans. Oil Temp OTP Trip	PCS Off	R	0	65535	0
				1	Trans. Oil Press Fault					
				2	Trans. Oil Level Fault					
				3	Trans. Low volt Temp OTP					
				4	FAN1 OCP					
				5	FAN1 UCP					
				6	FAN2 OCP					
				7	FAN2 UCP					
				8	FAN3 OCP					
				9	FAN3 UCP					
				10	FAN4 OCP					
				11	FAN4 UCP					
				12	FAN5 OCP					
				13	FAN5 UCP					
				14	FAN6 OCP					
				15	FAN6 UCP					
7663	30307	Uint16	SYS_FLT_YBOARD_2	0	AC SPD1 Fault	PCS Off	R	0	65535	0
				1	AC SPD2 Fault					
				2	Trans High Volt Switch feedback fault					
				3	Trans Amb. Temp. OTP					
				4	Trans Amb. Temp. UTP					
				5	FUSE Cabinet-A Amb. Temp. OTP					
				6	FUSE Cabinet-B Amb. Temp. OTP					
				7	Switch Cabinet-A Amb. Temp. OTP					
				8	Switch Cabinet-B Amb. Temp. OTP					
				9	Y Board Temp OTP					
				10	Cabinet water inlet Fault					
				11	Switch Cabinet-A Amb. Temp. UTP					
				12	Switch Cabinet-B Amb. Temp. UTP					
				13	System Altitude High Fault					

				14	Trans HV Switch close fault					
				15	Reserved	NA				
7664-7669	30308-30313	Uint16	NA		Reserved	NA	R	0	65535	0
766A	30314	Uint16	SYS_WARN_1	0	Trans. Oil Temp. OTP Warning	Warning	R	0	65535	0
				1	Cabinet High Humidity Warning					
				2	Fiber Comm. Loss Warning					
				3	System Parameter Mismatching					
				4	Module Number Loss Warning					
				5	System Power Mismatch					
				6	DC SPD Warning					
				7	Module Id config Mismatching					
				8	Config Two Master DSP Warning					
				9	Master DSP SPI Comm Lost Warning					
				10	Auxiliary Trans1 Temp Warning					
				11	Auxiliary Trans2 Temp Warning					
				12	Module HeartBeat Warning					
				13	Module CheckSum Warning					
				14	Central controller Ethernet Comm. Loss fault					
				15	Module IMD Check Failed Warning					
766B	30315	Uint16	SYS_WARN_2	0	Trans Oil Level Warning	Warning	R	0	65535	0
				1	Trans Oil Press Warning					
				2	Rsv					

7.2. Module Alarm

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	Bit	DESCRIPTION	Action	RW	Min	Max	Default
Power Module #1 Start Address: 31000 Length: 10										
7918	31000	Uint16	POW_MOD1_FLT_AC_GRID	0	RS Over Voltage(RMS)	PCS Off	R	0	65535	0
				1	ST Over Voltage(RMS)					
				2	TR Over Voltage(RMS)					
				3	RS Under Voltage(RMS)					
				4	ST Under Voltage(RMS)					
				5	TR Under Voltage(RMS)					
				6	AC Capacitor Rn Over Voltage(RMS)					
				7	AC Capacitor Sn Over Voltage(RMS)					
				8	AC Capacitor Tn Over Voltage(RMS)					
				9	Over Frequency					
				10	Under Frequency					
				11	Phase Sequence Fault					
				12	Phase Open Fault					
				13	Phase Lock Fault					
				14	Reserved	NA				
				15	Reserved	NA				
7919	31001	Uint16	POW_MOD1_FLT_AC_OUT	0	Module R Over Power	PCS Off	R	0	65535	0
				1	Module S Over Power					
				2	Module T Over Power					
				3	Module R Over Current(RMS)					
				4	Module S Over Current(RMS)					
				5	Module T Over Current(RMS)					
				6	Module Current Unbalance(RMS)					
				7	AC Capacitor R Over Current(RMS)					
				8	AC Capacitor S Over Current(RMS)					
				9	AC Capacitor T Over Current(RMS)					

				10	Reserved	NA				
				11	Reserved	NA				
				12	Reserved	NA				
				13	Reserved	NA				
				14	Reserved	NA				
				15	Reserved	NA				
791A	31002	Uint16	POW_MOD1_FLT_DC_SIDE	0	DC Input Voltage UVP	PCS Off	R	0	65535	0
				1	DC Input Voltage OVP					
				2	DC Inverse Fault					
				3	DC Input Current OCP					
				4	Bus Sum OVP					
				5	Bus Sum UVP					
				6	Bus Capacitor OVP(1/4)					
				7	Bus Capacitor UVP(1/4)					
				8	Bus Half Unbalance					
				9	Reserved	NA				
				10	Reserved	NA				
				11	Reserved	NA				
				12	Reserved	NA				
				13	Reserved	NA				
				14	Reserved	NA				
				15	Reserved	NA				
791B	31003	Uint16	POW_MOD1_FLT_CON_INV	0	INV1 R Over Current(RMS)	PCS Off	R	0	65535	0
				1	INV1 S Over Current(RMS)					
				2	INV1 T Over Current(RMS)					
				3	INV2 R Over Current(RMS)					
				4	INV2 S Over Current(RMS)					
				5	INV2 T Over Current(RMS)					
				6	INV1 Current Unbalance(RMS)					
				7	INV2 Current Unbalance(RMS)					
				8	INV1/2 Current Unbalance(RMS)					
				9	INV1 R Over Power					
				10	INV1 S Over Power					
				11	INV1 T Over Power					
				12	INV2 R Over Power					
				13	INV2 S Over Power					
				14	INV2 T Over Power					
				15	INV1/2 Instantaneous OCP					
791C	31004	Uint16		0	DSP System Disable	PCS Off	R	0	65535	0

			POW_MOD1_FLT_HARDWARE	1	FPGA Hardware Fault					
				2	FPGA Write Fault					
				3	FPGA DSP Ready Fault					
				4	FPGA EPO Fault(from Emif)					
				5	FanI1 Fail					
				6	FanI2 Fail					
				7	FanO1 Fail					
				8	FanO2 Fail					
				9	FanO3 Fail					
				10	FanO4 Fail					
				11	FanO5 Fail					
				12	Inlet OTP Fault					
				13	FPGA IGBT Short Circuit Fault					
				14	SPD WARN	Warning				
				15	Reserved	NA				
791D	31005	Uint16	POW_MOD1_FLT_COMM	0	TCP with System Loss	PCS Off	R	0	65535	0
				1	EMIF with SysDsp Loss					
				2	EMIF with mFPGA Loss					
				3	Fiber with SysDsp Loss					
				4	RTC Fail					
				5	Flash Fail					
				6	Fram Fail					
				7	Reserved	NA				
				8	Reserved	NA				
				9	Reserved	NA				
				10	Reserved	NA				
				11	IMD Check Time Out	PCS Off				
				12	Module Sent Fail	PCS Off				
				13	Reserved	NA				
				14	Reserved	NA				
				15	Reserved	NA				
791E	31006	Uint16	POW_MOD1_FLT_FPGA	0	FPGA System Disable (from IO)	PCS Off	R	0	65535	0
				1	AC OCP Fault					
				2	BUS OVP Fault					
				3	DC OVP Fault					
				4	DC OCP fault					
				5	P+12V OVP/UVF Fault					
				6	C Board OTP Fault					
				7	IGBT OTP Fault					
				8	A Board OTP Fault					

				9	INV1/2 Inductance L OTP Fault					
				10	INV1 IR CBC Fault					
				11	INV1 IS CBC Fault					
				12	INV1 IT CBC Fault					
				13	INV2 IR CBC Fault					
				14	INV2 IS CBC Fault					
				15	INV2 IT CBC Fault					
791F	31007	Uint16	POW_MOD1_FLT_BATTERY		Reserved	NA	R	0	65535	0
				0	FW Ambient OTP					
				1	FW Ambient UTP					
				2	FW IGBT OTP					
				3	IGBT Temperature Unbalance	PCS Off				
				4	DCI OCP(Reserved)					
				5	DCU OVP(Reserved)					
				6	IMD Low Fault					
7920	31008	Uint16	POW_MOD1_FLT_OTHER	7	IMD Low Warning	Warning	R	0	65535	0
				8	RCM High Fault (Reserved)					
				9	AC Preload Fault					
				10	Bus Preload Fault					
				11	Bus Soft-start Fault					
				12	ADC Calibration Fault					
				13	IGBT UTP	PCS Off				
				14	Reserved	NA				
				15	Reserved	NA				
				0	DC Preload Relay1 Fail					
				1	DC Main Relay1 Fail					
				2	DC Main Relay2 Fail					
				3	AC Preload Relay1 Fail	PCS Off				
				4	AC Preload Relay2 Fail					
				5	AC Main Relay1 Fail					
				6	AC Main Relay2 Fail					
				7	AC Main Relay3 Fail					
7921	31009	Uint16	POW_MOD1_FLT_RELAY	8	DC main relay1 warn	Warning	R	0	65535	0
				9	DC main relay2 warn	Warning				
				10	Reserved	NA				
				11	Reserved	NA				
				12	Reserved	NA				
				13	Reserved	NA				
				14	Reserved	NA				
				15	Reserved	NA				

Power Module #2 Start Address: 31100	Length:10
Power Module #3 Start Address: 31200	Length:10
Power Module #4 Start Address: 31300	Length:10
Power Module #5 Start Address: 31400	Length:10
Power Module #6 Start Address: 31500	Length:10
Power Module #7 Start Address: 31600	Length:10
Power Module #8 Start Address: 31700	Length:10
Power Module #9 Start Address: 31800	Length:10
Power Module #10 Start Address: 31900	Length:10
Power Module #11 Start Address: 32000	Length:10
Power Module #12 Start Address: 32100	Length:10

8. System Information

8.1. SN & App Version

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
7530	30000	Uint16	OS_DB_VERSION	Database Version	NA	R	0	65535	0
7531	30001	String	OS_SN_SEC_1	String System Serial Number 1. Example:0x'5741' = 'WA'	NA	R	0	65535	22337
7532	30002	String	OS_SN_SEC_2	String System Serial Number 2. Example:0x'4D4A' = 'MJ'	NA	R	0	65535	19786
7533	30003	String	OS_SN_SEC_3	String System Serial Number 3. Example:0x'3030' = '00'	NA	R	0	65535	12336
7534	30004	String	OS_SN_SEC_4	String System Serial Number 4. Example:0x'3030' = '00'	NA	R	0	65535	12336
7535	30005	String	OS_SN_SEC_5	String System Serial Number 5. Example:0x'3030' = '00'	NA	R	0	65535	12336
7536	30006	String	OS_SN_SEC_6	String System Serial Number 6. Example:0x'3030' = '00'	NA	R	0	65535	12336
7537	30007	String	OS_SN_SEC_7	String System Serial Number 7. Example:0x'2020' = ' ',	NA	R	0	65535	8224
7538	30008	String	OS_SN_SEC_8	String System Serial Number 8. Example:0x'2020' = ' ',	NA	R	0	65535	8224
7539	30009	Uint16	OS_FW_APP_VER_1	Center control unit App Version 1	NA	R	0	65535	0
753A	30010	Uint16	OS_FW_APP_VER_2	Center control unit App Version 2	NA	R	0	65535	0
753B	30011	Uint16	OS_FW_APP_VER_3	Center control unit App Version 3	NA	R	0	65535	0
753C	30012	Uint16	OS_FW_APP_VER_4	Center control unit App Version 4	NA	R	0	65535	0
753D	30013	Uint16	OS_FW_APP_VER_5	Center control unit App Version 5	NA	R	0	65535	0
753E	30014	Uint16	SYS_DB_VERSION	Sub-system database version	NA	R	0	65535	0
753F	30015	Uint16	SYS_FW_APP_VER_1	Sub-system App Version 1	NA	R	0	65535	0
7540	30016	Uint16	SYS_FW_APP_VER_2	Sub-system App Version 2	NA	R	0	65535	0
7541	30017	Uint16	SYS_FW_APP_VER_3	Sub-system App Version 3	NA	R	0	65535	0
7542	30018	Uint16	SYS_FW_APP_VER_4	Sub-system App Version 4	NA	R	0	65535	0
7543	30019	Uint16	SYS_FW_APP_VER_5	Sub-system App Version 5	NA	R	0	65535	0
7544	30020	Uint16	SYS_FW_FPGA_VER_1	Sub-system FPGA Version 1	NA	R	0	65535	0
7545	30021	Uint16	SYS_FW_FPGA_VER_2	Sub-system FPGA Version 2	NA	R	0	65535	0
7546	30022	Uint16	SYS_FW_FPGA_VER_3	Sub-system FPGA Version 3	NA	R	0	65535	0

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8.2. System Time Information

ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
7550	30032	Uint16	OS_TIME_YEAR	Local Time: Year	NA	R	0	9999	2022
7551	30033	Uint16	OS_TIME_MONTH	Local Time: Month	NA	R	0	12	1
7552	30034	Uint16	OS_TIME_DAY	Local Time: Day	NA	R	0	31	1
7553	30035	Uint16	OS_TIME_HOUR	Local Time: Hour	NA	R	0	24	0
7554	30036	Uint16	OS_TIME_MIN	Local Time: Minute	NA	R	0	60	0
7555	30037	Uint16	OS_TIME_SEC	Local Time: Second	NA	R	0	60	0